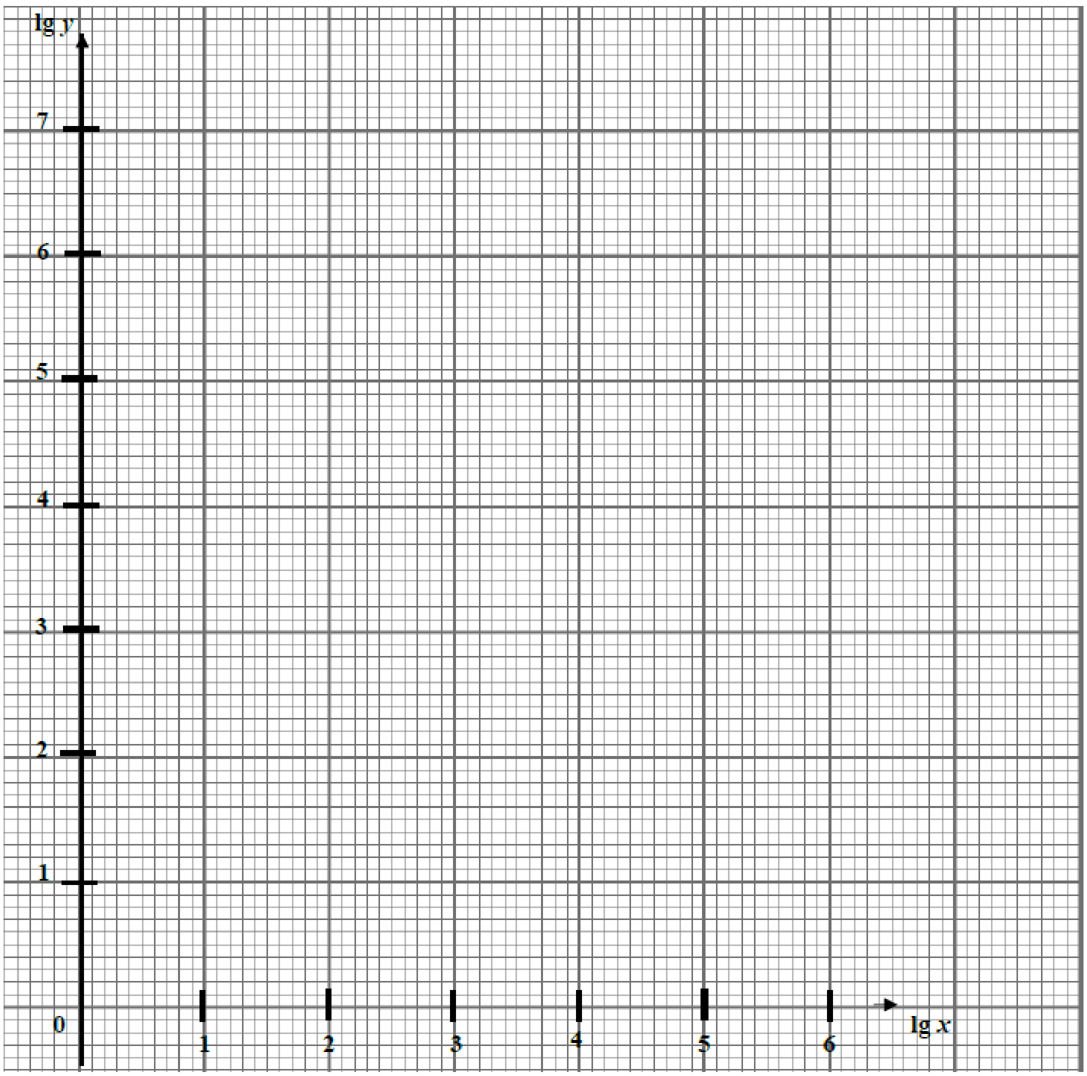
1	**ACSBR/2025/I/14**

The table shows the frequencies, y kilohertz, of a radio station at various wavelengths x metres.

x (m)	252	1500	10000	31623
y (kHz)	1190	200	31.6	10

It is known that a relationship of the form $y = px^n$ exists between x and y, where p and n are constants.

(a) On the grid below, plot lg y against lg x and draw a straight line graph.



(b) Use your graph to estimate the value of p and of n. [4]

(c) By drawing a suitable straight line on your graph, solve the equation $px^n = 10^{3.5}$. [2]

2 **BASS/2025/I/02**

In a biology experiment, a scientist is studying the growth of a bacterial culture under specific conditions.

The number of bacteria y, after t hours is believed to follow an exponential growth model given by the equation $y = Ke^{-nt}$ where K is the initial quantity of bacteria and n is a constant.

Measurements of t and y are shown in the table below.

t	1	2	4	6
y	243 000	49 000	2 000	82.3

(a) Plot $\ln y$ against t and draw a straight line graph to illustrate the information. [2]

(b) Using the graph, estimate the values of K and n. [4]

(c) By drawing a suitable straight line, solve $e^{t(1+n)} = K$.

State the equation of the straight line to be drawn. [3]

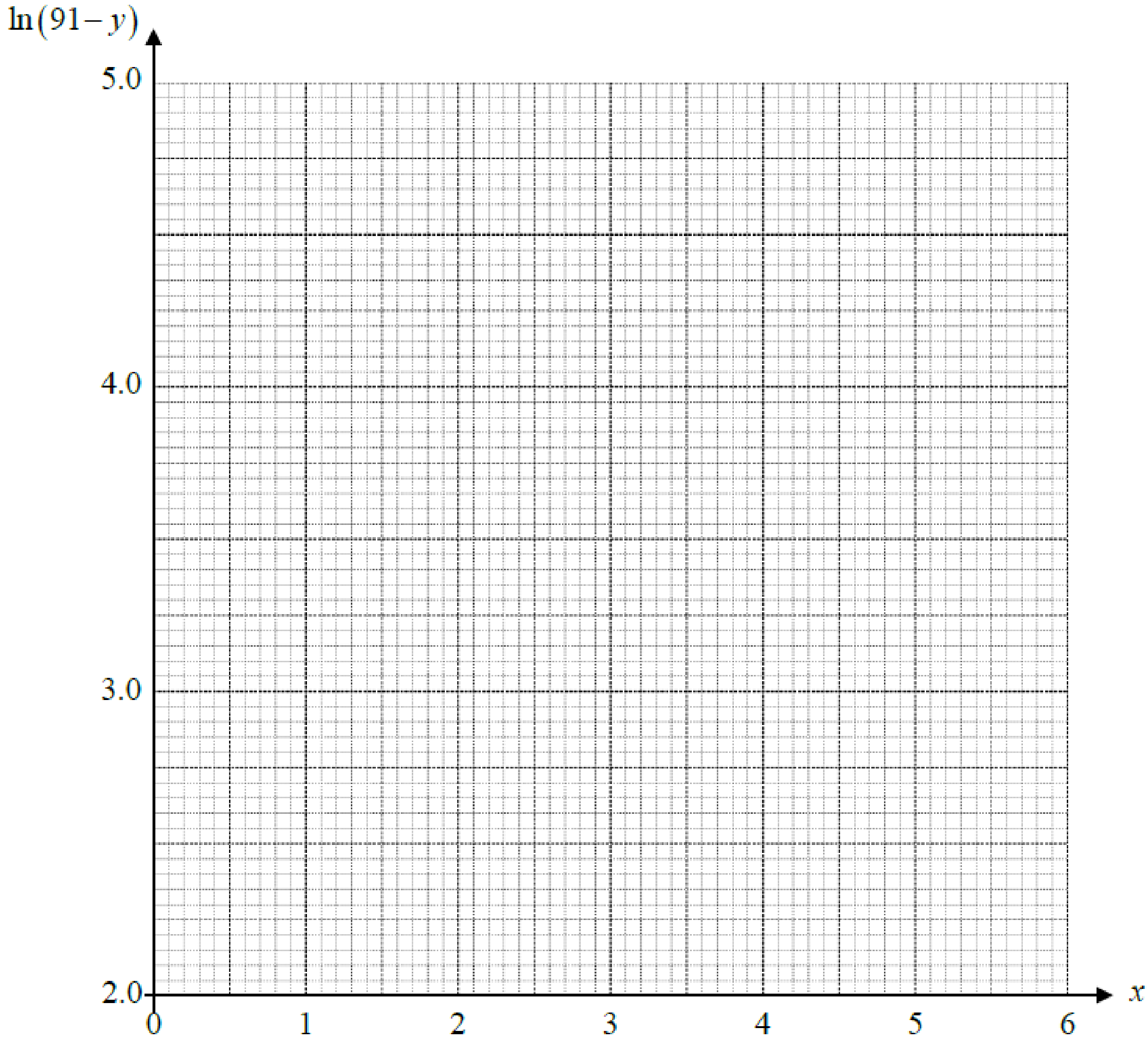
3 **CCHY/2025/II/8**

The table shows Andy's marks for his Mathematics practice papers each week.

Week x	1	2	3	4	5	6
Marks y	45	59	68	74	71	82

He believed that these figures can be modelled by the formula $y = 91 - Ae^{kx}$, where A and k are constants, by excluding one datapoint that does not follow the trend.

(a) On the grid below, by ignoring the datapoint that does not follow the trend, plot $\ln(91-y)$ against x and draw a straight line graph. [2]



(b) Use the graph to estimate the value of each of the constants A and k. [5]

(c) Suggest a possible reason why one of the marks does not seem to follow the trend. [1]

(d) From your straight line graph, estimate the expected marks for the datapoint that was excluded. [2]

4	GMS/2025/II/8

The mass, m mg, of a radioactive substance decreases with time, t hours.
Measured values of m and t are given in the following table.

t (hours)	2	4	6	8	10
m (mg)	48.2	41.5	35.7	30.7	26.5

It is known that m and t are related by the equation $m = m_0 e^{-kt}$, where m_o and k are constants.

(a) On the given grid, plot $\ln m$ against t for the given data and draw a straight line graph, using a scale of 1 cm for 1 unit on the t-axis and 2 cm to 0.1 unit on the $\ln m$ axis.
Form a linear equation of the line that you have drawn. [4]

(grid given on the next page)

(b) Using your graph,

(i) estimate the value of k and of m_o, [3]

(ii) estimate the number of hours for the mass of the substance to be halved. [2]

1

(a)

$\lg x$	2.4	3.2	4	4.5
$\lg y$	3.1	2.3	1.5	1

(b)

$y = px^n$

$\lg y = \lg p + \lg x^n$

$\lg y = n \log x + \lg p$

$n = -\dfrac{5.3}{5.5} = -0.964$

$\lg p = 5.5$

$p = 10^{5.5} = 316227 = 316000$ (to 3 sig fig)

(c)

$px^n = 3.5$

$\lg p + n \lg x = \lg 10^{3.5}$

$\lg y = 3.5$

$\lg x = 2$

$x = 100$

2

2a

Refer to graph paper
Points plotted correctly with correct scale + label axes
Best fit Straight line joining points

2b

x	1	2	4	6
$\ln y$	12.4	10.8	7.6	4.41

$y = Ke^{-nt}$

$\ln y = \ln Ke^{-nt}$

$\ln y = \ln K + \ln e^{-nt}$

$\ln y = \ln K - nt \ln e^{-nt}$

$\ln y = \ln K - nt$

$-n = gradient = -1.6$

$n = 1.6$

$\ln K = vertical \text{ int } ercept = 14$

$K = 1200000$ (3sf)

2c

$e^{t(1+n)} = K$

$\ln e^{t(1+n)} = \ln K$

$t(1+n) = \ln K$

$\ln K = t + tn$

$t = \ln K - nt$

$t = \ln y$

$t = 5.4 \pm 0.1$

3

(a)

(b)

Transforming the equation:

$$y = 91 - Ae^{kx}$$

$$91 - y = Ae^{kx}$$

$$\ln(91 - y) = \ln A + \ln e^{kx}$$

$$\ln(91 - y) = \ln A + kx$$

Using 2 points on the line (1, 3.8) and (3.5, 3),

$$\text{gradient} = \frac{3.8-3}{1-3.5} = -0.32$$

Reading off the Y-intercept, we have 4.15

The equation of line is

$$\ln(91 - y) = -0.32x + 4.15$$

By comparing,

$\ln A = 4.15$ $\qquad\qquad$ $k = -0.32$

$A = 63.4$ (3 s.f.)

(c)

Any reason within the context

- The lower mark on week 5 was likely to be due to a harder paper.
- Andy might have been ill.

(d)

The datapoint is (5, 71)

Based on the straight line graph, the reading should be 2.5.

The expected marks is

$$\ln(91 - y) = 2.5$$

$$y = 79$$

4

(a)

$$m = m_0 e^{-kt}$$

$$\ln m = \ln m_0 - kt$$

$$\ln m = -k(t) + \ln m_0$$

t (hours)	2	4	6	8	10
m (mg)	48.2	41.5	35.7	30.7	26.5
$\ln m$	3.88	3.73	3.58	3.42	3.28

(b)

(i) estimate the value of k and m_o,

$$\ln m_o = 4.03 \quad \Rightarrow \quad m_o = 56.3$$

$$\text{Gradient} = \frac{3.28 - 3.88}{-8}$$

$$-k = -\frac{3}{40}$$

$$k = 0.075$$

(c)

	$\frac{1}{2}m_o = 28.15$ $\ln(28.15) = 3.3369$ $t = 9.2$